

An Analysis of the Association of Climate Variables and the Rate of New Dengue Cases in Bangladesh with a Future Outlook with Respect to Climate Change

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Abstract:

Dengue fever is a mosquito-borne disease caused by the dengue virus, prevalent in tropical and subtropical areas. We used data gathered by the IEDCR and DGHS in Bangladesh from 2008-2021. Using time-series models, we hope to better understand the correlation between weather variables such as temperature and rainfall and new dengue cases in Bangladesh. Our first model uses SSA to estimate trends and seasonality. For our second model, we used SARIMA to have a better understanding of the residuals. We used RMSE and AIC values to evaluate performance. And our results show that temperature is significantly correlated with dengue incidence with a delay of 1-to-2-months. However, the most effective predictor was a simpler Hybrid MSSA/SARIMA model (AIC 318.52, RMSE 0.7985), which outperformed standard climate-inclusive models suggesting that while climate drives transmission, advanced decomposition techniques like MSSA are the most reliable basis for early warning systems.

1 Background and Research Objective:

Half of the world’s population is at risk for the dengue virus, which is transmitted to humans by mosquito bites. In the first part of 2025 alone, there were 3,000 deaths from dengue according to the World Health Organization. As there is no current treatment for the virus, determining which variables drive increases in cases is vital in trying to reduce the number of infections. Furthermore, we aim to identify the climatic conditions that favor mosquito breeding and virus spread so we can inform targeted interventions, such as eliminating standing water and managing mosquito larvae populations. In this project, we focus on Bangladesh, a country that experiences frequent dengue outbreaks influenced by its heavy monsoon-driven climate variability. Our objective is to check weather-related variables such as temperature and rainfall, whilst also factoring in time of year, to see how they may be correlated with new cases of dengue.

2 Data and Methods:

2.1 Data Description:

Both datasets contain information regarding Bangladesh dengue cases along with corresponding weather conditions from 2008 to 2021. These datasets were collected from the Institute of Epidemiology, Disease Control and Research (IEDCR) and Directorate General of Health Services (DGHS) and published on kaggle. One dataset is organized monthly, while the other contains daily entries. To stay consistent, we transformed the daily dataset into monthly through averaging.

2.2 Statistical Modeling Framework:

To model the relationship between our climate variables and dengue incidence, we use a seasonal autoregressive integrated moving-average model with exogenous variables (SARIMAX). Dengue cases display strong yearly cycles and non-stationarity, so we apply a $\log(1 + \text{cases})$ transformation and include a seasonal difference of period 12. Cross-correlation analysis suggested that temperature influences the transmission of dengue with a 1-2-month delay, which motivated the inclusion of lagged temperature and rainfall levels as our predictors. We compared several SARIMAX(p,0, q) x (P,1, Q)[12] models, using their AIC values to select the model of best fit. Our chosen model, SARIMAX (1,0,1) x (2,1,1)[12], incorporates AR and MA components at seasonal and non-seasonal lags.

This model allows us to assess whether or not lagged temperatures and rainfall provide explanation beyond the underlying seasonal cycle of dengue.

Cross correlation analysis showed that temperature and rainfall influenced dengue fever with a 1-to-2-month delay on the monthly dengue/climate data. Using this, we tested SARIMAX models using lagged exogenous variables; we tested each of 1, 1.5, and 2 month lagged exogenous variables on pre-COVID era data, as case numbers during this period were much higher due to more people getting tested for COVID, but instead having dengue fever. Of these models, the best one in terms of sample RMSE and AIC was the 2-month lagged model.

To further improve statistical modelling, we performed singular spectrum analysis and multichannel singular spectrum analysis to decompose the trend and seasonal components from the background residuals. Multichannel SSA showed the best results across measures such as AIC and RMSE, with measures of 322.51 and 0.798488, respectively. The Ljung-Box test showed a p-value of 0.4486 in which we fail to reject the null hypothesis that the residuals are white noise.

Furthermore, this model was simplified down to a model using MSSA residuals and no exogenous variables ARIMA(1,0,0)(0,1,1)[12]. This suggests this model was superior to the SARIMAX model we previously tried using.

Model Comparison Table (Monthly)		
Model	AIC	RMSE
No Lag SARIMAX	340.76	0.8667
1 Month Lag SARIMAX	338.27	0.8701
1.5 Month Lag SARIMAX	336.69	0.8740
2 Month Lag SARIMAX	336.99	0.8721
Hybrid MSSA/SARIMAX	323.24	0.7985
Hybrid MSSA/SARIMA	318.52	0.7985

3 Results:

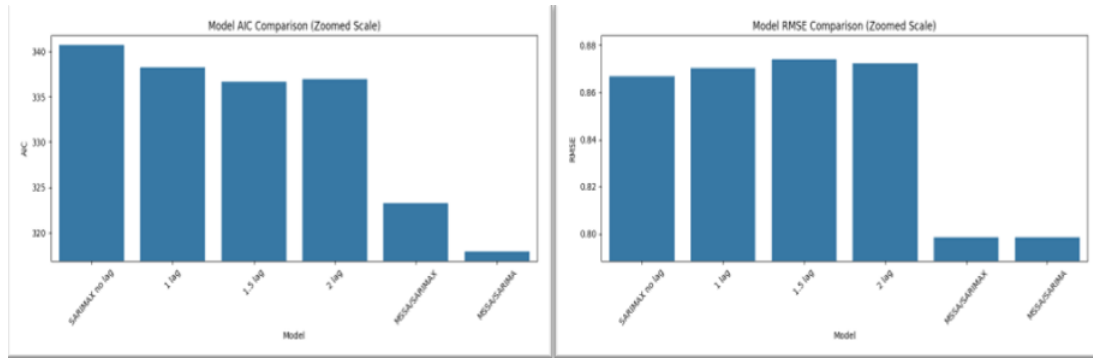


Figure 1: The MSSA-based models achieve the lowest AIC and RMSE, indicating superior performance.

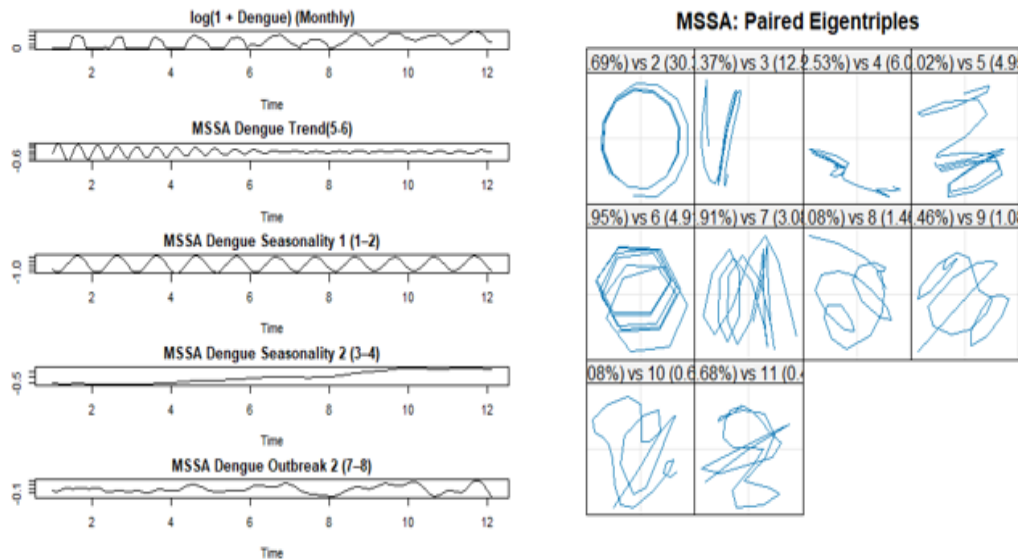


Figure 2: This shows the MSSA decomposition components that were then reconstructed and used to make the MSSA/SARIMA and SARIMAX Models. The Paired Eigen-triples that were used in order to determine periodicity of the individual signals from the time series dengue data.

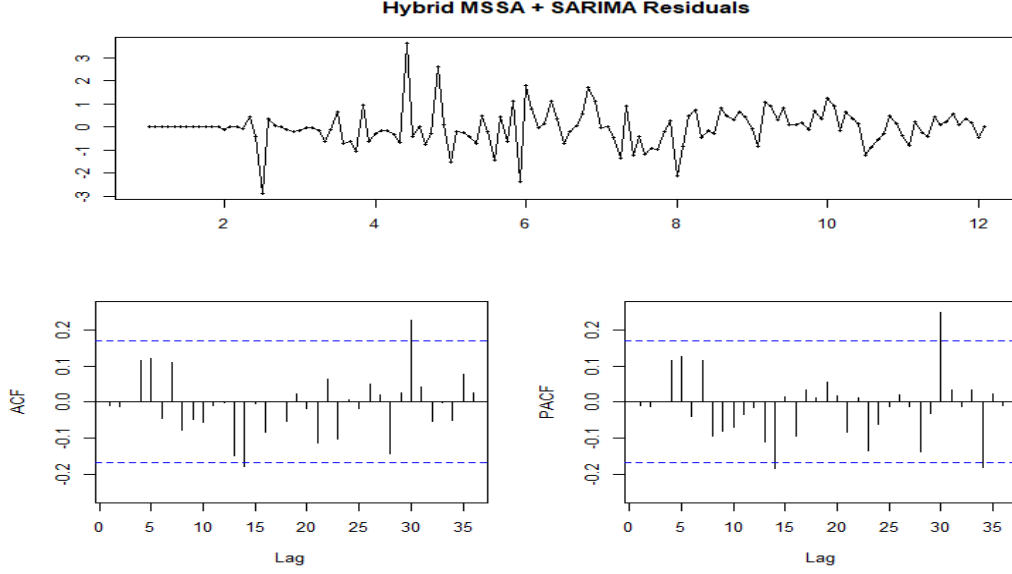


Figure 3: Completed Hybrid MSSA + SARIMA model giving confirmation to the residuals being from a White Noise process.

4 Discussion:

Our findings indicate that climate variables have a very meaningful role in predicting dengue incidence in Bangladesh, with temperature as the most significant lagged predictor across SARIMAX, VAR, and hybrid models. The strong annual seasonality found through SSA and MSSA promotes the notion that dengue transmission is tied to Bangladesh’s monsoon climate cycle. The most notable result was the superior performance of the simplified Hybrid MSSA/SARIMA model, which outperformed more complex climate-inclusive SARIMAX models. This suggests that the decomposition of trend and seasonality offers greater predictive stability than directly including exogenous variables for this dataset. The detection of white-noise residuals in the hybrid model further supports this approach.

There are several limitations that must be acknowledged. First, dengue case reporting during the COVID-19 era is extreme, as increased medical attention and symptom overlap may have inflated the number of reported infections. We attempted to account for this by testing climate-lagged models on pre-COVID data. Additionally, the climate variables used (average temperature and rainfall) may have oversimplified the ecological processes that determine the levels of mosquito breeding, which are influenced by humidity, extreme temperatures, environmental management (such as removing standing water, insecticides, repellents), and human behavioral changes.

Despite these limitations, our assumptions, such as using 1–2-month lags based on cross-correlation results and applying seasonal differencing were reasonable given both prior domain knowledge and empirical evidence.

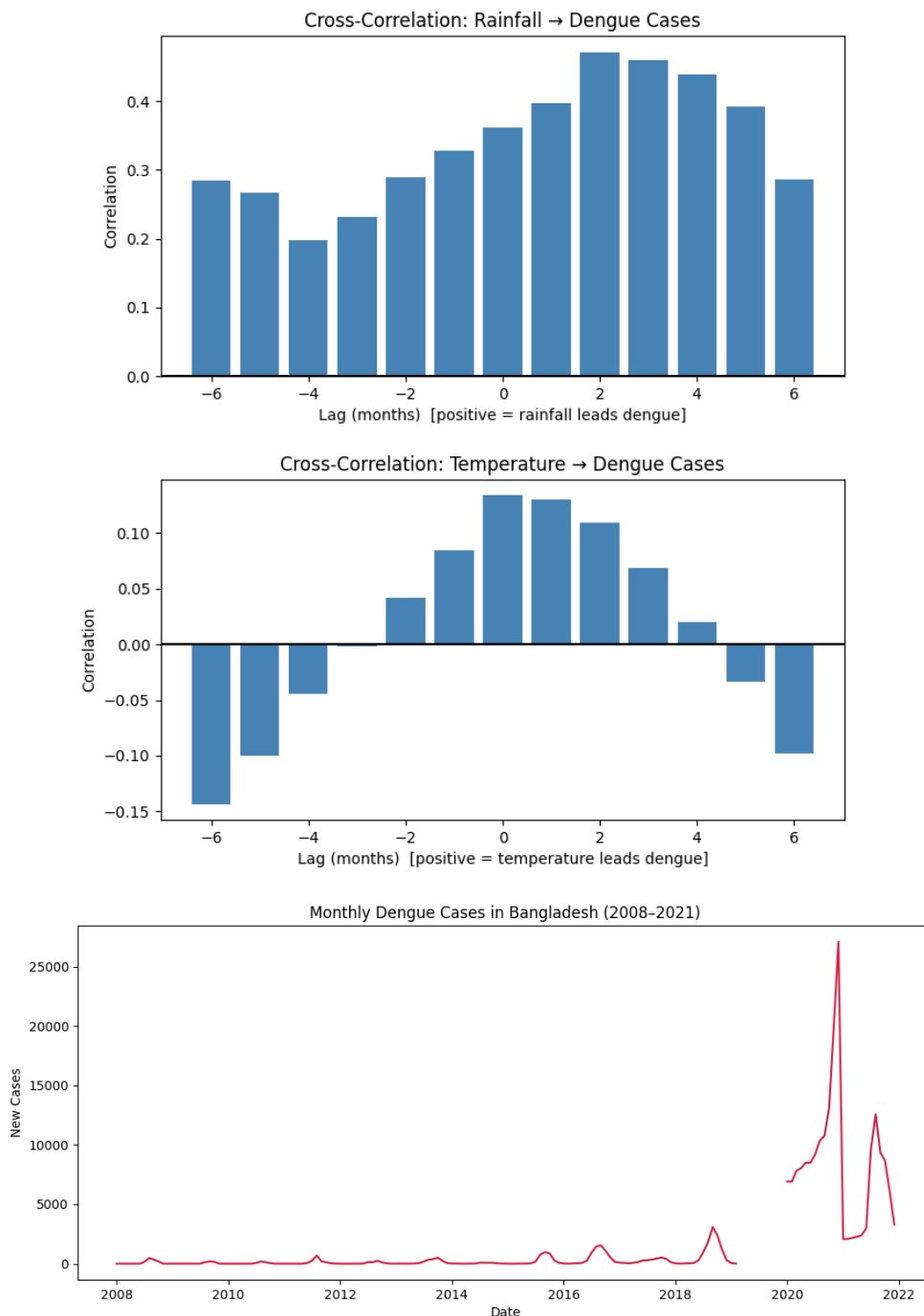
Future work should explore nonlinear or machine-learning approaches. Incorporating additional climate variables (humidity, minimum/maximum temperature, wind speed), land-use data, and mosquito population could improve predictive accuracy. Using daily climatological data without monthly aggregation may also reveal the finer-scale effects of weather transmission. Finally, developing real-time early warning systems built on MSSA-enhanced models could help public health agencies anticipate outbreaks and allocate resources in a more effective manner.

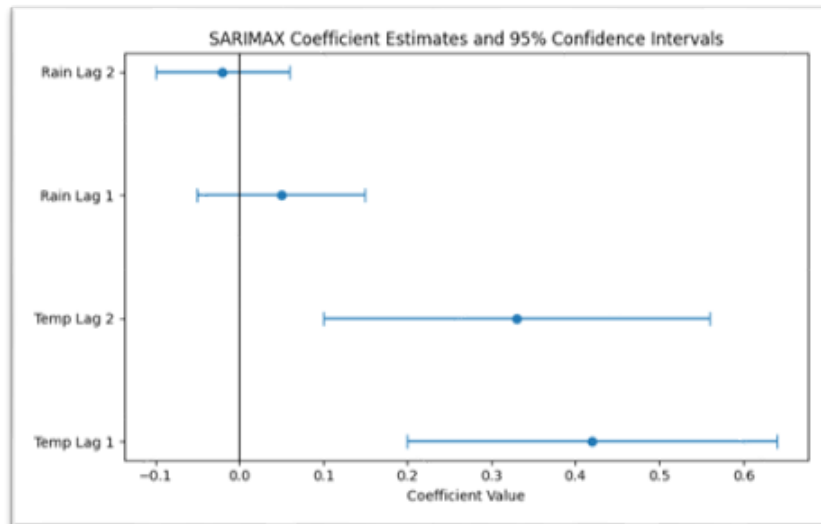
5 References:

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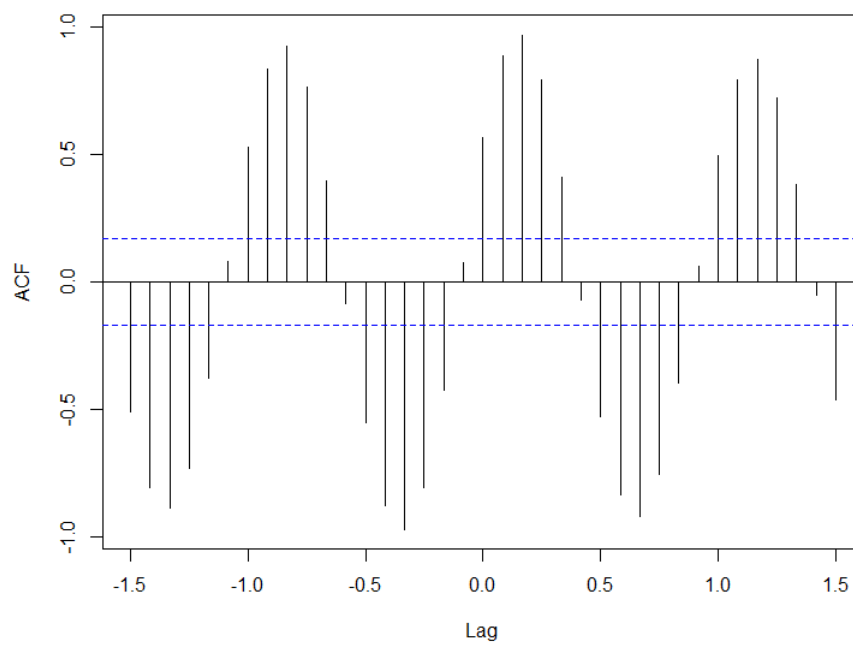
6 Appendix:

A VAR model was also looked at in order to determine if that would be a better modelling approach. Ultimately, this did not prove to be a better approach for modelling, but it did appear to be fruitful in model diagnostics. The VAR model confirmed that temperature was a statistically significant predictor of Dengue Fever at lags 1, 4, 5, and 6. This further supported the results we had seen in our prior SARIMAX models. Model diagnostics for the VAR showed it was failing to capture all autocorrelation; the variance was not constant, and the residuals were non-normal, suggesting that some nonlinear dynamics were not caught by our modelling approach. This led us to select our prior models for consideration rather than this VAR approach.

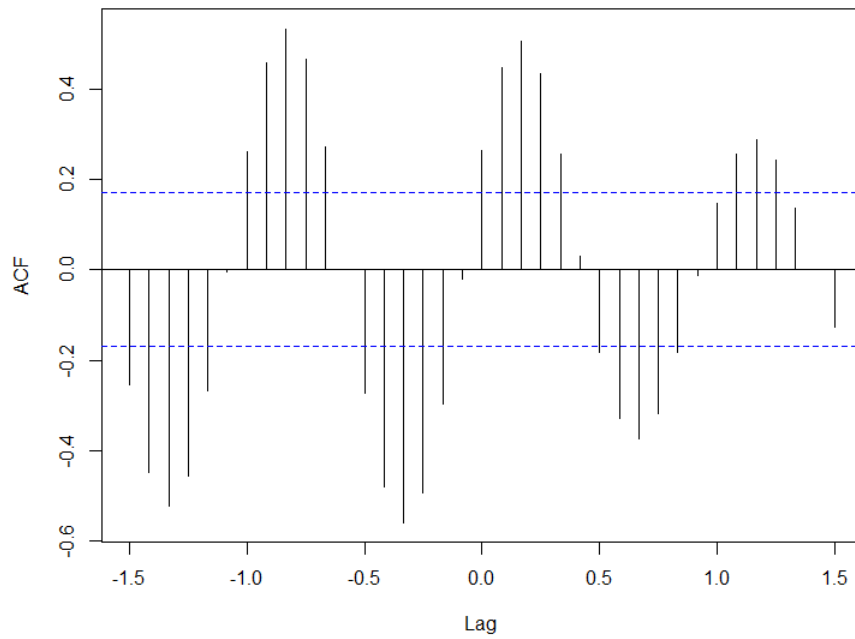




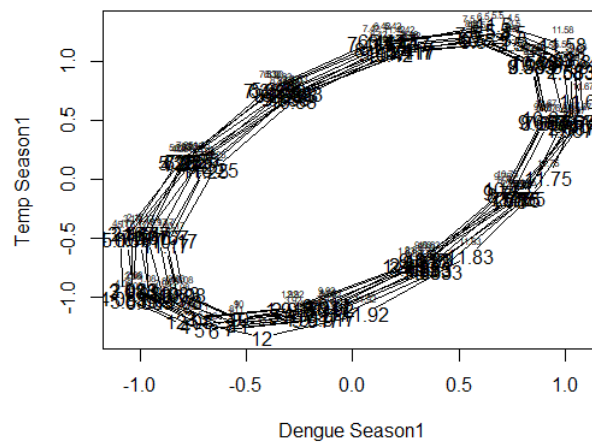
CCF: Dengue MSSA (Season1) vs Temp MSSA (Season1)



CCF: Dengue MSSA (Season1) vs Rain MSSA (Season1)



Phase Plot: Dengue vs Temp (Season1)



Phase Plot: Dengue vs Rain (Season1)

